

Not in my Backyard!? Lessons from a Community Conflict

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ABSTRACT

We present a case study of a local community debate about a land development proposal. The debate was not amicably resolved. It entrained an encampment, a lawsuit, a local electoral upheaval, and sustained outrage. It was a serious community conflict. Through a set of converging interviews and field visits with key actors from throughout the community, researchers reconstructed what happened, gathered diverse views about how and why it happened, and reviewed the shorter- and longer-term outcomes of the conflict. As can be true in significant conflicts, all participants can be seen as rational and well-intentioned, and some of the outcomes are clearly positive for the community. But the cost was high in terms of civility, trust, and community coherence. Implications include both insights into, and opportunities to re-conceive and redesign, local governance and planning policies, practices, and information support.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**; **Empirical studies in collaborative and social computing**;

KEYWORDS

community conflict, community engagement, information infrastructure, policy

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1 INTRODUCTION

Conflict in communities have existed since their inception. Conflicts such as disagreements and differences in opinions arise due to heterogeneity in a community. Conflicts are a pivotal part of community development because it rekindles concerns and connections towards local community, it highlights issues that are important for the community, and gives voice to the heterogeneous

opinions and perspectives [12]. Coleman famously described the dynamics of community conflicts by comparing and contrasting different community cases. In his book, Coleman [10] explains that community conflicts tend to progress in a downward spiral, resulting in polarization, weakened relations, and the disruption of equilibrium. In the context of community informatics and human-computer interaction, studying community conflicts is beneficial as it helps in carrying out important community-based work and create solutions to address conflicts and strengthen a community. With communities becoming more networked and driven by information infrastructures, civic participation correspondingly is becoming increasingly data and information driven. Hence, it is important for design research to study community conflict in the information age [6].

This research delves into one such case study that was driven by concerns over water crisis and urban sprawl, and an ensuing conflict between a township and a university. Before the crisis, the township of State College and Penn State University had a healthy relationship. However, this relationship become tenuous after a proposal for the development of a university-owned land in the township. This could potentially cause storm-water run off to nearby wells which are primary sources of drinking water in the area. While some community members have opposed the land development due to its eventual impact on water quality, the anti-development movement has also played a pivotal role in the backdrop. Thus, the land development proposal served as a trigger, resulting in a widespread community conflict, which subsequently led to the emergence of evidence-based activism and supporting infrastructure to tackle the problem. This study gives a narrative of what happened in State College, the perspectives of key stakeholders, the background of environmental and municipal government regulations that constrained the narrative, and reflections and lessons learned from the process.

The conflict presented here can be construed as a “not in my backyard” phenomenon, described in policy and planning literature as the NIMBY sentiment. The NIMBY sentiment is the community response to unwanted land development [42]. Local residents often acknowledge the importance or need of the proposed planned development, but not near their neighborhood [11]. NIMBY sentiment is usually stimulated by health, environmental or social implications the land development might have on the community. Although NIMBY usually has a negative connotation to it, where it can usually sour community relations, it can also have positive outcomes for the community, such as bringing missed development factors and views into light. The actors involved in this community conflict, irrespective of their affiliation, followed laws, regulations and proper procedure to take actions, but that turned out to be insufficient in preventing a conflict. Against this background, this

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paper seeks to answer the following research questions: *How do people make community conflict better or worse?*

We describe an episode of community conflict that has implications for information infrastructure for planning, deliberation, and communication between multiple stakeholders, what citizens and local government can do to improve outcomes within the current regulatory regimes, implications for how laws and other regulations might be changed to better encourage and leverage citizen participation. Our primary contribution is identifying lessons learned by the various actors in this case study

- Early community engagement in development processes is crucial irrespective of whether it is required by law to include stakeholders at an early stage or not.
- Design and dissemination of information to the community in a comprehensible and engaging manner.
- Preservation and knowledge management of historical context is important in maintaining transparency and awareness.

Even though this paper is a singular case study of a community conflict, the insights and implications drawn from this research can serve as a useful example and a guide for managing similar future conflicts.

2 RELATED WORK

To contextualize the research setting and our insights, we present a review of core conceptual phenomenon as it relates to communities. These include community engagement, citizen participation and efficacy. We also provide a review of empirical and design based research and applications in the civic technology space.

2.1 Community Engagement, Participation, and Conflict

The concept of "community" has been extensively studied in various disciplines. Societal groups bounded by a geographical area can be called a community when the people in that local space feel connected to each other and care about each other [15]. Gusfield [20] distinguished between two characterizations of a community. One focusing on the geographical and the second focusing of the relational. Numerous studies have been conducted to examine community processes using these dimensions. In addition, Carroll and Rosson [8] developed a conceptual model of community that comprises of three facets - community identity (enables attachment), participation and awareness (enables engagement), and multiplexed support networks. In a democratic society, these facets of community, form an important factor and lead to community engagement and participation. Community engagement and participation leads to awareness of what is going on, who does what, and participation which arises from taking responsibility of circumstances [5]. This strengthens the relationship between citizens and their local government, which in-turn leads to a stronger democratic society.

Arnstein [2] was perhaps the first researcher to conceptualize the citizen participation in a local community. He used the metaphor of a ladder to convey the interplay of power distribution and participation in a community. The higher the individual is on a ladder, the more power the person has to influence decision-making. Hence,

participation without redistribution of power seems to be meaningless. Citizens can be actively engaged with the local government, but if they don't have the resources or power to influence change and decision-making process, this participation and engagement is meaningless.

Community identity, participation, and engagement all determine strength of a community. However, these facets have a counter-productive effect where they can sometimes give rise to community conflicts. For conflicts to arise, a community needs to be engaged, aware and participate in democracy [10, 12]. Conflicts have been more extensively studied at an organizational level, where the focus is largely on determining whether a conflict would prove to be functional or dysfunctional for the organization [37]. Conflicts in an organization or a community can have positive consequences, negative consequences, or both, depending on the actors, resources, context, power and organization [18]. The consequences can create adaptive and integrative effects for the community, and also unravel power structures and power differences, disrupting the social structure of the community. In design research community, conflict is often analyzed from the lens of a political perspective called agonistic pluralism. This perspective entails the value that conflicts are innate to any democratic society, and heterogeneous opinions should be channeled towards a positive social change [34]. Design of infrastructure in such a society can be a critical tool in shifting the conflict towards agonism [23]. Hence, design is both a tool and a process that can lead to social and political change through exploration, participation, and deliberation of different stakeholders and artefacts.

2.2 Community Collective Efficacy

Community engagement with respect to various issues and challenges is enabled by beliefs that the community collective is capable, that it has specific capacities. Such beliefs were originally theorized by Bandura [3] as self-efficacy, which is an individual's belief of their capacity to participate in a domain. This work was further extended to the concept of collective efficacy which was a group's collective capacity to participate in a domain. Community collective efficacy was further developed as a specialization of collective efficacy [7]. Community collective efficacy beliefs are predictive of and instrumental to positive community outcomes [40]. They are also a pathway to collective action tendencies [46, 47]. Carroll et al. [9] developed a community collective efficacy scale with 17 items. The scale explains capacity analysis of general issues any community might face. Items on these scale reflect on how communities can manage conflicts by creating capacities, community relationships, and active community involvement. In the case presented here, believing that the community can protect the quality of local source water despite potential threats from development makes it rational and more feasible to organize and exert efforts that indeed secure water quality.

2.3 Civic technology & Digital Governance

There is a slew of scholarly work in the broad area to engage citizens in local issues like voting [27], planning [31] and deliberations [45]. Research has taken creative forms by leveraging public displays to generate awareness and increase participation in physical space [41,

43]. Similarly, activism is also a topic that is emerging within HCI [33, 39]. We see further value in re-conceptualisation of activism as evidence-based [38] refocuses our fixation from protest and violence to co-production of knowledge.

The topic of civic engagement has also been attractive for practitioners. For example, there are umpteen examples of mainstream technology platforms that promote citizen engagement and participation. Neighborhood forums like Nextdoor, crowdfunding platforms for civic and or local issues like Kiva MicroFunds, platforms for citizen-government communications like SeeClickFix and Change.org are some pervasively used social participation platforms. Social media sites like Facebook facilitate social interactions, mobilize resources [28] and developing social capital [1]. Lately, they have also been promote neighborhood events and community happenings both by local citizens, municipalities and businesses alike.

Although not exhaustive, it can be seen from the extant work, there is rising interest in the cross-disciplinary research agenda of Digital Civics [48]. The goal in this paper is to extend this strand of research through the lens of community conflict.

3 METHOD

This research used a case study approach to understand and investigate several layers in the conflict. Yin highlights the importance of case study as an empirical strategy that weaves together diverse forms of evidence for an entity, individual or unit of analysis [50]. The case of community conflict presented in this paper is a particular episode of land-development related community dispute. Hence, it required in-depth understanding of the context of conflict and its implications on communities and design. Even though its a single case study, the insights and lessons from this research can still be applied to future similar contexts, because this kind of land-development conflict in a small university town in United States is not atypical.

As reverberated by several case study researchers [14, 35, 50], we used multiple sources of evidences in the form of semi-structured in-depth interviews with 8 participants, reviewed existing documents including analog and digital traces relevant to land sale, local news and media coverage and completed two critical field visit to the activist encampment, and final community meeting after the conflict was over. During the first field visit, researchers met with community members who were camping at the site and protesting the land sale and development. During the second field visit, researchers attended a community meeting at a local church where different community groups had gathered to discuss future scenarios of land-use management and preservation. Data from interviews was audio recorded by the consent of the participants, and data from field visits was recorded as notes taken by the researchers through observations and conversations. The audio recorded data was transcribed and qualitatively analyzed along with field visit notes. We used open coding to extract important elements and concepts from the data [49]. We first used axial coding to conduct a divergent analysis where we did preliminary examination, comparison and categorization of data. Next, we used selective coding to conduct a convergent analysis, where these preliminary categories were further examined, compared and finally categorized into three main

themes: *actions taken by different community groups, perspectives and beliefs of different community groups, and reflections and lessons learned from the conflict*. This study was reviewed and approved by the researchers' university institutional review board.

Findings were synthesized using an embedded case study design that included both individual people as well as broad-based systems as units of analyses. For interviews, individuals were chosen who well-represented parties concerned in this particular land sale-development socio-political network related to the land development episode. A brief description of the interviewee demographics is given in Table 1.

Participant ID	Gender	Representation
P1	F	Community Member
P2	M	Community Member
P3	F	Community Member
P4	M	University Official
P5	M	University Official
P6	M	University Official
P7	F	Township Official
P8	M	Local Water Authority Official

Table 1: Interviewee Demographics

4 STUDY CONTEXT

State College is in a historic town and gown relationship with Penn State University. This has been a matter of pride, stemming from fruitful inter-dependence and exchange between community members and university expertise. However, this relationship became tenuous when a proposal was introduced for the development of university-owned land in State College.

A local non-profit foundation had divested thousands of acres in State College roughly fifteen years ago. A large part of that parcel was gifted to Penn State University, with an agreement that most of that acreage would be re-purposed for public use, both passive and recreational. At the time it was outside the regional growth boundary of State College. Subsequently, it was rezoned as a planned residential development in 2004 by the State College supervisors. Roughly a decade after the rezoning decision, the university sold a part of that parcel to a local land developer, who intended to develop high density student housing. At the same time, the State College wanted to build a parkland, so they reached out to the university for help. The university worked with the regional council of governments and selected a parcel right next to the land developer's development site, for development of the regional park. Several events occurred from the time a huge amount of acreage was acquired by the university to the time when the land developer laid down plans for a proposed development deal.

Local community members were unaware that the university had sold a land parcel adjacent to the proposed regional park to a land developer for construction of high density student housing. They were concerned that the resulting construction would have a severe effect on regional water sources and degrade the scenic beauty of the area. These events spanned almost fifteen years. Community members became involved in the debate in the last three

years, raising concerns about construction initiating stormwater run off to nearby drinking water wells that were primary sources of water in the area. Members of the local community worked together to oppose the land development. They feared that in addition to subsequent risks of water contamination, this development would be the onset of urban sprawl. In spite of a lawsuit and multiple appeals, community members were not successful in their effort to halt the land sale. While proposing exit routes to this conflict, community members demanded a land swap by Penn State University to the land developer. This was also constrained by real estate law, which compels contract execution (i.e. selling the land parcel) and prohibits backing out of either parties in a transaction.

Even though the community members failed to halt the development, their myriad efforts throughout this process induced several lessons for different community members, and also led to other positive incidental outcomes. Community members were now more aware of different laws and processes, university and government officials learned the importance of early and more efficient engagement of community in such processes, and the various efforts led by the community helped raise \$2.75 million for a local non-profit, to buy 400 acres of that land for conservation easement. This conservation easement was planned to be developed into a recreational hiking and biking area. The land development proposal served as an exogenous shock to the community, much like a natural experiment context to study the resulting community conflict, as well as how evidence-based activism emerged and progressed, providing implications for information infrastructure support in this space.

5 FINDINGS

The researchers identified three main actors in this case study: the community residents, university officials, and civic officials. Each of these actors were either supporting or opposing the land development. The findings from data analysis provides insight into the how and what kind of actions were taken by different actors, different perspectives and how those influenced the community, and reflections and lessons learned by these actors from this process.

5.1 Actions Taken by Different Community Actors

The community members opposing the land development did extensive research to collect pertinent information about the proposed development including its history, review prior studies on the impact to water quality, and understand numerous rules, regulations and ordinances about zoning, rezoning, land development, and township decisions. P7 enthusiastically shared her data collection endeavours:

"I have a lot of documents if you want the detail. It's like you are learning so much all at once. You are doing all this work, all this effort...I have done a lot of deed research. I am no expert on it and its complicated...But when I put things together it made sense." (P7)

P7 further explained how the construction of the park on an adjacent land was tied together to the proposed housing development plan. This, according to P7, was going to seed ever more development and have dire consequences on the water quality in the nearby wells. She was worried that there would not be a sustainable land use.

"If you read the documents carefully, you will notice a binding contract to develop this road that is access to the park...This is evidence of intention of a domino effect. There is no need to test the engineering of a road if you didn't intend to build it." (P7)

Other motivated community members saw this as an opportunity to take a stance and make their voices heard. Decisions to act were also fueled by the "mute spectator" role that other non profits and water related agencies assumed. Community members formed an activist group to represent the interest of the community. For several months, groups of activist members were encamped at the disputed land site. The activists were a mix of people from diverse backgrounds, including scientists and conventional farmers. They reached out to several people to get an understanding of the context.

"We built networks and bridges with agencies to get all the documents." (P3)

The researchers met with some of the members of the activist group just as they were getting evicted from their occupation. In these site interviews, the activist group members explained their position to oppose the development by the land developer in a very detailed and articulate manner. The leaders struck us as being highly technically engaged. They knew a lot about the history of the site, the underground geology, the surface streams, the surrounding land, possible interactions and risks in storm-water scenarios. As P1 explained, their roles in collecting and synthesizing relevant data were considered as being 'community hawks'. During the site visit, she mentioned that they were building a database to store all the collected documents from different agencies, deeds, and other water-related data.

"Sara has created a database with all these documents.. She is also a reporter and a community builder..We as a group did a significant amount of research...She was very instrumental in sending out all the information including the links to actual documents so people could do their own interpretation. Very often elected officials do not get to do that. They get a brief distillation and believe it or not a decision is made and it is over." (P1)

The conflict became apparent and more visible to the public when the activist group tried to file a lawsuit against the land developer. However, activists were constrained by the regulatory complexity involved in filing a lawsuit. Community members had thirty days to synthesize their preliminary research and present a case, in court, to halt the proposed development. P2 explained that 30 days was not enough time for them to take action:

"While filing a [lawsuit], you need to establish both a basis for objection and legal standing - legal standing means convincing a judge that you personally will be harmed by the planned development. The objection itself is a problem in that the developer and landowner had years to make their case, but you have just 30 days." (P2)

The lawsuit was filed based on a protecting technicality that there was a five acre parcel which was not part of the original planning document, was going to be used for stormwater management. The initial ruling by a local judge was in favor of the activists, however an appeal made by the land developer later overturned the decision. Even though the activists lost the lawsuit, the legal process caused a two year delay in the development process, thereby creating an opportunity for local citizens to become more aware of the land development process.

On the other hand, the university officials in charge of the land development process explained the actions they took once they saw the opposition from the community. The officials acknowledged the limitations in current community engagement processes, and how lack of proactive engagement and dissemination of information to the public caused a community conflict.

“You know, not a lot of people go to, you go to a local government meeting, there’s like five people. It’s not they don’t draw crowds most of the time to the average Monday night local government meeting. So by the time somebody had found out that this was, you know, probably on a path that was, it was very far along the path, it was relatively too late.” (P5)

In addition to the local township meetings, local government had its own email lists that they used to disseminate information. P7 recounted that she first got involved with the land developer’s development process when she signed up for this email list.

“When I would move to a different municipality, I would check out their website and sign up for their email notices. So I mostly ignored them. I just couldn’t look at them. I didn’t know really what they were saying, fine. But then I was doing that. And I saw in it, it was in all caps. I swear that’s what, like drew it to me. And I looked at it and when I read what it said, I was like, what!?!..and I thought to myself, that makes no sense to me because there’s nothing out there, that’s like, wildlife area. And I thought, there’s no way they’re doing it out there. I know something about it is protected.” (P7)

P7’s knowledge and interest in township email lists was because of her personal history. She grew up in a family of activists, and had always been interested in staying actively involved in local community events and developments. She got involved and initiated the push back against the land developer before she became an elected official of the municipality. Hence, even though the development was a public process in the eyes of the government, very few citizens were aware of it because of the inefficient communication channels that failed to engage the community. P7’s involvement was an exception because of her special background and interests.

As traditional attempts to inform the community failed and the conflict started to brew, university officials attempted to synthesize and distribute information to the community. They created a document detailing the timeline of series of events in the process of this transaction. This document was meant for public utilization and was distributed to the community through various sources such as local newspapers.

“I have a one pager and I can print out for you, that we developed for community, yeah, for public consumption...It took us a month to get this one pager together to make sure we had all of our facts, right, that has the series of events, the history in some narrative” (P5)

The documentation contained information describing real-estate transactions between Penn State University (seller) and other buyers, its timeline and an aerial image of the land parcel. Going over the timeline, P4 explained that the land parcel was originally donated to the university roughly fifteen years ago. At the time it was outside the regional growth boundary of State College. Subsequently, it was rezoned as a planned residential development in 2004 by the State College supervisors. Roughly a decade after the rezoning decision, the university decided to sell the land to the land developer with the intent of developing high density residential housing. Building on this sequence, university officials, P5 and P6,

elaborated on the timeline shared by P4. They described how the parcel of land came into possession by the University because a trust divested in the area and gave large acreage to the university. P5 explained their first action steps in the proposed land development site and adjoining areas:

“It is part of the engagement with the trust, we agreed that we would re-purpose a lot of that acreage for public use like recreation. We worked with several municipalities to convey land to the [parks and recreation department in the area] to develop a regional park. State College came to us and asked for our help to facilitate and support development of a park to meet community needs. Over the course, we conveyed land to the water authority that is a quasi public agency and then gave some to the land developer.” (P5, male)

University officials also arranged several meetings with the members of the opposition groups in an attempt to come up with a consensus. The community members suggested alternative sites owned by Penn State University for the student housing development, however, the university had their own campus development plans for these sites, and hence could not negotiate for a land swap. In addition, they were bound by the real estate law that compels contract execution and hence they could not back out the transaction.

“There’s a piece of real estate law...called specific perform, that once we are in an agreement with a developer, they can force us to sell we can’t back out of this. So if we try to cancel the agreement of sale, the developer could have sued us. We would have had to pay lawyers fees, court costs, and we would have lost. I mean, you know, our counsel basically says you can’t, this is very clear. We have a signed agreement of sale. So at that point, we could not we couldn’t back out of it, single-handedly, we could not do that on our own.” (P4)

P4 explained how lack of feasibility of alternate sites suggested by the community members, and regulations of the real estate law did not leave any other option for the university officials. By this time, the community members had filed a court case against the development plan, but according to the university it was too late and at this point they could not back out of the deal and transaction with the land developer.

To summarize, the land development proposal prompted several community members to partake in, learn and act on the situation. This propelled university and civic officials to respond and take action by preparing documents, holding meetings, and engaging in public discussions to resolve the brewing community conflict.

5.2 Different Perspectives of Actors

The community members who were opposing the land development were primarily concerned with the effect on local water resources and degradation of the view shed. While the community members leading the opposition were basing their argument primarily on water resources concerns, a lot of other community members were initially drawn to the opposition because of their concerns about the degradation of the view, but later learned and got engaged with water quality issues as well. The opposing group of citizens in the community had formed a coalition group to oppose the development. P3 explains how she got involved with the opposing group in the community:

“And I’ve lived in State College for 38 years and I have a lot of interest and concern in the development that’s occurring around here

and the sudden extreme growth, pushing out into the green areas surrounding the township that's happening. So my entry into the coalition and into the whole issue with the land developer started when I realized that they were going to be basically ruining that amazing vista and secondary to that I began learning about big issues with the water quality." (P3)

Hence, joining the coalition turned out to be an opportunity for community members like P3 to learn about possible threats to the hydro-geology of the area and strengthen their case. P2 also shared similar sentiments as P3, where he was concerned about degradation of the view and impact on water quality. He lamented that it was unfortunate that there couldn't be data about the quality of the natural view. Hence, even though concerns over water quality affected everyone, degradation of the natural view also distressed the local citizens who shared strong community sentiments and attachments. These citizens felt that it was not just the quality of water that was being affected but also their quality of life was getting disrupted.

Another salient attribute of this conflict was that there were discrepancies about the quality and fidelity of the reviews. Permitting agencies and local expert hydrologists, whom community members referred to as an in-group, reviewed perceived threats to water quality. Community members were skeptical of the fairness of the reviews because they were examined by university hydrologists. Hence, lack of trust in provenance of information and reviews on water quality was a major cause of the conflict. In addition, when community members sought to review the technical reports themselves, it became very overwhelming to do so in a short thirty day time period. This was the time given to the community to synthesize and make their case before the court. Furthermore, there were limited interactions with other third-party watershed stakeholders about the impact analysis of the development. As P8, water authority official, mentioned

"We did not want to take a position. It was not our job to. However, we provided some recommendations for the storm-water management process to the proposed development." (P8)

Lack of trust in the university and government officials involved in the land development process stemmed from the lack of initial community engagement in the decision making process. The community members were upset and felt their rights were violated when the university sold the land for development without engaging them in the decision making process. Even though the land was owned by the university, the area was an integral part of the township and the residents felt a sense of belonging and ownership to this land. They felt that the historical town and gown relationship with the university was sabotaged. In addition, flaws in the communication strategies and channels also led to speculations by the community members about role of local government in passing various bills in favor of the development. P7 speculates how the land developer's bill must have passed right before she got elected:

"I'm quite sure that the board at that time, because you know you're elected in November and then you don't go in until January, the fitting board still makes legislation. That's critical time in there. They can do a lie if they want to. They passed the land developer's final plan in that period. Yes. And they passed it in an extremely incomplete manner. They did not, when they saw the turnovers they did not want that to collapse."

According to P7, the critical time between November and January, when the elected board of officials changes, was the time used to pass the land developer's development bill. The elected officials at the time knew the environmental impact and that the development requirements weren't met, and they wanted to finish the process before the next team of elected officials took over. This unsystematic process didn't leave time for any due diligence. According to P7 and other opposing members of the community, the reason for extending the regional growth boundary to accommodate student housing development was to just make profit out of that land,

On the other hand, University officials, although empathetic towards community sentiments, had their own perspectives on the argument made by the community opposition group. After numerous failed meetings with the members of the coalition and presenting water quality review reports from hydrology experts to them, the university officials realized that the members of the coalition were not going to trust the evidence presented to them because it came from the university experts. This mistrust became frustrating for the university because on one hand the coalition wouldn't trust the data and policy interpretation of the experts, and on the other hand they could not get third-party water stakeholders such as non-profit groups and water authority to provide another neutral point of view.

"But, you know, I really never got the sense that they really fully respected what the experts were saying. Or that they had their own interpretation of it. And they, they were trying to, you know, obviously represent their opinion as to what all that means" (P5)

P5 acknowledged the citizens had a right to voice their opinion, but the continuous failed efforts to negotiate made him question and point to other developments in the region which were seemingly much closer to the public wellheads and much more sensitive than the current site in question. He didn't understand why the opposing group did not debate these developments and were fixated on the issue of this student housing development transaction. Hence, he perceived the opposing group's interpretation of information and reasons for debate to be incorrect. This led to the speculation that the opposing group of community members are simply anti-development and did not want to deal the challenges of having student housing in their neighborhood.

"From my perspective, this was a group of local residents who did not want student housing in the neighborhood and the associated challenges that come with that. To some degree noise, to some degree traffic, to some degree, building on what is now a beautiful piece of land. So There was some degradation of the view shed, right, they didn't want their views to be obstructed by housing, and, but it was not truly about water quality. The arguments raised about the water were more in order to get greater community interest and involvement in this project." (P4)

P4 speculated that the issues of water quality raised by the opposing group were simply to get more traction and community interest in this movement. According to him and other university officials, the reviews from hydrologists did not indicate any threat to local water quality. The expert reviews said that any development in this region has the potential for challenges because of the nature of the karst formation. So any development in the region had the risk and possibility of creating sinkholes and ridges. Hence, this presented no greater threat than any other development in the region.

Moreover, the land developer was bounded by the law to comply and follow all local, state, and federal laws and regulations as it pertained to the various aspects of the proposed development plan. P6 explained how the land developer had followed all the prevailing statutes, ordinances and regulations regarding this initiative development plan. It was the developer's responsibility to address any issue such as stormwater and water quality, to get approvals from regulatory agencies and the local jurisdiction. These regulatory agencies, local jurisdiction, and the municipality were subject matter experts who could not approve any plans without reviewing all the pertaining issues.

"So throughout the entire process, our stated position was that the land developer has to comply and follow any and all local, state and federal laws and regulations as it pertains to the various aspects of the proposed development plan." (P6)

P6 explained how their perspective was that as long as the land developer followed all laws and regulations, they were within their rights to develop student housing on that property, and they tried conveying this to the opposing group as well. Further, a university official reasoned as to why other third-party nonprofits and stakeholders in the watershed refused to weigh-in in the debate:

"They didn't want to take a stance in this issue otherwise it would backlash. But that signaled to us that all the reviews were acceptable and right." (P4)

The supporting group felt that the debate became contentious which discouraged neutral third-party organizations such as nonprofits from taking a stance. Hence, mistrust and lack of effective communication led to failed negotiations, which in-turn turned the community conflict into a controversial debate. Even though the supporting group acknowledged the sentiments of the community residents, various confinements of the law restricted the actions the supporting group could take.

To summarize, different actors had different perspectives on the transaction, based on their affiliations and influences. Different affiliations and its effects lead to varying opinions, growing mistrust, and speculations within the community. These different perspectives of actors shaped the community conflict and made it more controversial.

5.3 Reflections and Lessons Learned

Both the supporting and opposing groups reflected on the whole process and debate and contemplated some important lessons learned from this conflict. These reflections led to insights both for researchers and the community as to how everyone can proactively take measures to avoid a future conflict on any community issue. All actors involved in the conflict, whether supporting or opposing the development, did not have harmful intentions for the community. The opposing group was not subversive; and neither was it the case that the supporting group were trying to undermine community concerns. Ultimately, everybody wanted fruitful development and community harmony.

First, there was an acknowledgement of inadequate communication and visibility in the land development process. From the community members perspective, developers and landowners control all of the planning and testing activity on the front end of

projects like this. They can work for years building up a thick portfolio of plans and results, expert testimony, etc. Then, they take it to the municipality, who are non experts and can get relatively quick action. P1, a community member vented her frustration in the process:

"...This is not known, visible to the public, you need to pay attention to the newspaper, every tiny notice/ article in it, pay attention to township agendas and attend all public hearings." (P1)

Assenting to the problem of visibility, P3 noted:

"Who knew Penn State was marketing [the land] to developers? Nobody knew. We were not aware. We had to find out through the right to know filing." (P3)

P3 believed that it was important for citizens to be proactive, learn about their rights, and take charge of situations. It was important for citizens to pay attention to local news, be aware of different processes and actions taken by local jurisdiction and other big authorities in the community, and instead of believing and blindly following the institution, do research and critically take actions. Hence, using the right to know laws, building networks and bridges with various agencies, and using the democratic system to change the makeup of boards and supervisors, are some of the actions that the opposing group took which can be enhanced in the future.

Second, one of the most insightful reflection from members of the supporting group was how the history of the land in question got lost in time and debate. A non-profit foundation gifted the land parcel to Penn State University for recreational use. They divested of property and conveyed parcels in and around the township region to the university for public use at the university discretion. The original transaction was aimed at supporting a regional park and bringing public recreation. The community through their elected officials, desired and stated through their actions by working with the township region government in pursuing public grants for public planning purposes. Hence, the story around the collaboration of the university with municipalities and regional authorities somewhat got lost. A university official noted several events had occurred before he had assumed his job:

"I was not even here when this happened, I have inherited this with this job. But we have created a timeline documenting the history of the land parcel." (P4)

The history of original land transaction got lost with time and the current officials involved in the process inherited the repercussions. This was also one reason why lack of knowledge of past transactions and processes, it became difficult to negotiate a solution during the community conflict.

Third important reflection and lesson learned was to improve ways to educate and inform passive residents of the community.

"Before you even get to how to interpret data, you need to understand what are the inherent rights of a property owner in proposing development in accordance with local zone. What is that process? What is the land development process? And I don't think public agencies do a well enough job. Do they do a good job of advertising hearings, but they don't do a good job explaining processes." (P5)

P5 noted that various laws and regulations pertaining to land development and zoning can be complicated and confusing to average citizens. Hence, it is important to find ways to communicate local ordinance and regulatory information in a comprehensible

way, so that residents remain aware and are on the same page with the authorities.

Fourth reflection and lesson learned was about engaging the community early on. Penn State University officials acknowledged the work that needed to be done to improve communication and engagement with the community members at earlier stages of the land development proposal process. As a university official, P4 was concerned about the friction that was built through this debate and recognized the place for an early community involvement in the process.

“At one point I was like it is a good thing. People are keeping us honest. Members of the community are keeping us accountable. And if we can’t answer the questions we shouldn’t be doing it...So we wanted to answer in an informed way...We created the timeline and we shared with media and put it on our website but, we were not so good at pushing it out.” (P4)

Even though the university officials thought they checked every box with respect to following the rules and communicating the process with the community, it was later in the conflict that they realized the impediments of the poor communication strategies used. The different layers of the municipal government and regulatory permitting agencies were mundane and citizens had to be proactive in attending meetings to understand and learn about land development guidelines, what issues regulatory agencies have purview over and what issues they don’t have purview over. It was recognized that besides improving communication methods, it was important to involve the community early on and have a more inclusive process with all the stakeholders.

5.3.1 The Aftermath. Even though the community members could not reach a resolution to end the conflict, the reflections and lessons learned led to insightful discussions and the community came together to reach a consensus. A local environmental non-profit organization raised \$2.75 million to purchase approximately 400 acres of the land for conservation easement. The non-profit partnered with Penn State University and local community groups to raise funds and develop plans for the purchased land. Penn State University’s landscape architecture students formed project teams to brainstorm and prototype different ideas for developing and maintaining the conservation easement land. Once the prototypes were ready, the community groups, non-profit organization and the university held a public meetings at a local church to exhibit student led ideas and projects. Different stakeholders at the meeting had discussions with each other and then reviewed the prototypes and plans for development.

The researchers attend the meeting as part of field observations for this study and spoke with different stakeholders. Figure 1 shows community members discussing one of the student project team’s work. The students spent two semesters carefully studying the land and its properties as part of an upper-level landscape architecture course. They developed different land-use scenarios and management strategies based on the course, and inputs from various stakeholders throughout the two semesters. The participants discussed these ideas with the student teams and offered feedback. After a series of meetings and discussions, the community identified seven consensus items and design elements for the conservation easement land site. These items included design elements to build



Figure 1: Public meeting to discuss possible land-use scenarios

and maintain hiking, biking and running trails for the community, build riparian buffers, and maintain existing forest land by promoting native plants and trees and removing invasive flora.

In addition, the coalition formed by the community members opposing the land development, later turned into an environmental protection group. Their focus became on educating the community about the local natural resources, possible threats to these resources, and what community residents could do to protect these resources. They collected information about local land water system history and current use and started creating information repositories on their website, they help public meetings to discuss local environmental issues, and collaborated with other local non-profit environmental community groups. Hence, even though it started as a coalition of activists opposing land development, the conflict turned the coalition into a community group learning and teaching each other about local land and water systems. This was a big step towards a more engaged, knowledgeable, and active community.

6 DISCUSSION

This paper delves into a case study of a community conflict between Penn State University and State College. Penn State University sold a university-owned land in State College to develop high-density student housing. This concerned the local citizens because such a development could potentially be detrimental to local water quality and an indication of urban sprawl. The groups supporting and opposing the land development, took several actions to raise their point of view and engage in discussions to reach an agreement. These actions such as community outreach, and research and collection of information, engendered different perspectives and beliefs that shaped the conflict and gave rise to many controversies. Ultimately, the community could not reach a consensus with respect to the land development dispute, and the land developer went ahead with the development of high-density student housing and a recreational park adjacent to the housing. However, looking back at the dispute, all community members reflected on the events that led to the conflict and shared some valuable lessons learned from the conflict.

6.1 How do people make conflicts better or worse?

In academic literature, community is often regarded as a core human social structure that must comprise facets such as community identity, participation and awareness, and multiplexed support networks [8]. These facets make up an ideal model of community that increases citizens' learning, attachment, and engagement towards their community. However, these facets can have a flip side to them as well, where increased citizen engagement wouldn't always mean that people are generally content and happy. Increased learning, engagement, and attachment can lead to citizens becoming more knowledgeable, opinionated, and liable demonstrators towards certain community decisions and policies. This flip side of the different facets of a community can lead to disputes and conflicts. Tenets of community conflict research regard it as an essential factor in continued community and organizational development [4, 12]. However, organizational research on conflict views it to have both positive and negative consequences for an organization [30, 37]. It has positive consequences if it promotes organizations core values such as productivity, adaptability, and stability, and negative consequences if it impedes these values.

This view can be applied to community conflict as well, where people make a conflict better if the controversies and disputes ultimately improve different community facets such as identity, participation, and awareness. In the case of conflict between Penn State University and State College, initial events and actions by both supporting and opposing group made the conflict worse and ultimately, the community could not reach an agreement that would please everyone. Lack of early community engagement and improper and late dissemination of information to the public fueled the conflict and caused mistrust between university and community residents. The university and civic officials used public town hall meetings during the early phases of the transaction that were largely ineffective in drawing and engaging citizens to community issues [26]. The reviews done by hydrologists to examine possible water quality issues at the land development site was not a public process until the very end when the dispute became more contentious. On the other hand, community members opposing the land development were reluctant in trusting the reports on inspection of water quality because the experts who conducted these reviews were university employees. The community member's lack of evidence on substantial water quality issues, made their case seem like a NIMBY sentiment where their only concern was degradation of the vista. In addition, lack of input and intervention from neutral third-party organizations made it difficult for groups to negotiate and trust each other.

However, the community also learned from each of their missteps which provided an opportunity for all groups and actors to come together and take a step towards conserving the land that was so important to the community's sentiments and the environment. They raised funds to purchase a part of the land that was up for sale for student housing, and held several community meetings to come up with land-use scenarios to preserve and manage the land. In addition to this land conservation, the entire episode sparked a new found interest and involvement in local water quality and other environmental issues. The coalition which was originally

formed by community residents to protest the land development, now turned into an environmental group, stimulating community interest and learning about their local watershed, and taking initiatives to protect and conserve their local water system. Hence, the aftermath where different actors realized the imbalance and disruption their dispute had caused in the community, brought them together to reflect on their missteps, and stride towards a more informed, amicable and engaged community.

6.2 Implications for Design

Design can be an important driver for both supporting and creating social and political change. Hence, design is both a tool to support social change, and a process itself to spark deliberation and exploration. Design research has adopted some critical theoretical and conceptual lenses such as speculative design, feminist utopian design, and participatory design to critique and analyze the process and implications of an artefact [21]. These theoretical and conceptual lenses help unpack roles and stakes of different participants, and how participation by each stakeholder influences change. For instance, Parra-Agudelo et al. [36] provide design directions for social change in developing communities using agnostic and adversarial design epistemology. The goal of design is not conflict resolution, but integration of multiple views and information to possibly engender positive change. Our research of community conflict in State College provides similar insights where we discovered how disengagement, lack of early communication, disintegrated and lost information led to a conflict that despite honest efforts and deliberations left all parties incapable of any resolution. In the light of these insights, we draw implications and explain three design directions.

First, historical background about the land, why it came to be, and the order of sale of neighboring parcels spans almost two decades. All parties involved in the conflict recognized that there were many facets to the development proposal and, furthermore, that the timeline for related decisions was lost with time. During this time, there were personnel changes in the township and the university. Details about what events occurred, what decisions were taken and when, and who made them, become relevant to understanding context and easing transition between individuals. An ecosystem of all data-in-place [45] that incorporates these details would aid in navigating and making sense of historical circumstances. Moreover, this would reduce the burden to retroactively reconstruct historical events and pave way for a more productive and rich deliberation among parties. In addition, knowledge management systems can be another important information systems to preserve historical context and information for communities. Knowledge management systems have been established as a crucial information system for both organizations [29] and for designers [44] to re-purpose data, information, and processes without re-inventing the wheel. Design that preserves data and information in a community context is important for growth and preservation of community institutions. In this case, historic institutions played a critical role in shaping the nature of the conflict. The burden of historically lost documents and decision-making had to be carried out by current institutional actors, entangling them in a dispute that became difficult to address

without access to historical contexts. Hence, making historical institutionism accessible will lead to creation of more rational and constructivist institutions [16].

Second, community members, while reflecting about their workflow and field actions, mentioned how they were meticulously involved in data collection and meaning-making of sequence of events pertaining to the land sale, rezoning, and storm-water management plan reviews. The researchers witnessed normal citizens become local activists. Their concerns extended beyond their own backyards. Additionally, their position in discussions and the overall debate was significantly data-driven in contrast to the personal and turf-protective sentiments espoused by NIMBYs. Some community members demonstrated evidence-based and informed activism. A place-based data ecosystem would further balance the distribution of knowledge and other resources. It could also serve as an incentive for constructive debate and discussions. Further, traditional planning and deliberation processes can be augmented with geographic information systems (GIS) such as those proposed in [17]. Matters of township, especially those having spatial attributes, including zoning, rezoning and land development could be deliberated using simple terminology such as land-resource indicators [22]. Data about a place belongs to the citizens of that community. Hence, data about land-development, zoning, water quality, and other regulations are all part of community data [6] that should be always be accessible and comprehensible to the citizens. Easy access and comprehension of community data can increase community capacity, community efficacy and engagement.

Third, this community conflict reveals information sharing practices that seem out of step with contemporary tools and methods. The township relied on institutionalized practices, which include posting notices in local newspapers about land sale transactions and solicitations of community input via public hearings. Many community members missed these notices and solicitations perhaps expecting that such things would be distributed through social media or some other channel. Paradoxically, as a society, we have become adept acquiring higher volumes of information than ever before. Yet, at the same time, there seem to be expectations that this information travels through particular channels. It did not occur to many of our interviewees that they would have to look outside of these particular channels to get in the loop. This reinforces a need for the local township to embrace other forms of media to share information and calls for participation [19, 24, 25]. One potential way to be cognizant and attentive to community attitudes and beliefs that are usually left out in planning is by collecting community narratives through video storytelling as in [32].

Community conflict in design research is often seen from the lens of a political perspective called agonism. DiSalvo [13] has done extensive work on adversarial design where he seeks to explore this political perspective through design research. Adversarial design is an inquiry into agonism. Disputes and conflicts in a democratic community are inevitable. Rather than channeling design work on finding solutions to conflicts, design research can offer a platform to shift the narrative towards a more agonistic perspective. This means creating pathways for individuals and institutions to have access to information, multiple viewpoints and an instrument to communicate and emerge from a conflict as a stronger community.

Conflicts are an inherent part of every community, and should be channeled towards community growth. Our design implications of State College conflict seek to provide design directions towards an agonistic perspective. Better access to historical information, improved communication channels, and integration and public access of community data, are three fundamental ways information and multiple viewpoints of this conflict can be made more visible and accessible to the community.

6.3 Policy Implications

Community members have thirty days to react to a decision and establish a legal standing to reverse a decision that is taken by a township. This is a rather narrow window for average citizens to mobilize, get a firm grasp on existing circumstances, scrutinize each earlier decision and present a coherent argument to appeal in the court. Allowing room for community engagement and input from the beginning is crucial to reduce costs incurred, minimize bitter feelings and strengthen trust. The small time frame currently offered to citizens was strikingly limiting, less accommodating and a barrier to enacting social change. An open process offers numerous opportunities for people to disagree, but nonetheless it seems to produce agreement at the end fairly reliably.

7 CONCLUSION

Community conflicts are not new. What is new is our understanding of them and the value they hold in giving us a fresh impetus to reflect and learn from them. Through meaningful engagements with all parties involved in a conflict about a proposed land development site by the land developer, researchers were able to witness first-hand the implications of lacking information infrastructure and communication. The contentious debate shed light on concerns about development in originally to be protected land, differences in interpretation of sequence of events and resentful feelings. The conflict narrative presented in this paper, is a case at hand that directs our attention to bringing reform in the way we plan and deliberate in communities. Information platforms hold immense power to mediate such community conflicts and facilitate sound decision making. It is in the best interest of all of us to achieve consensus in an informed, accessible and amicable way. For future work, it would be beneficial to study similar such community conflicts and develop richer understanding and theoretical framework. It would also be beneficial to conduct longitudinal and comparative studies to see how appropriate information systems can change and shape a community.

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